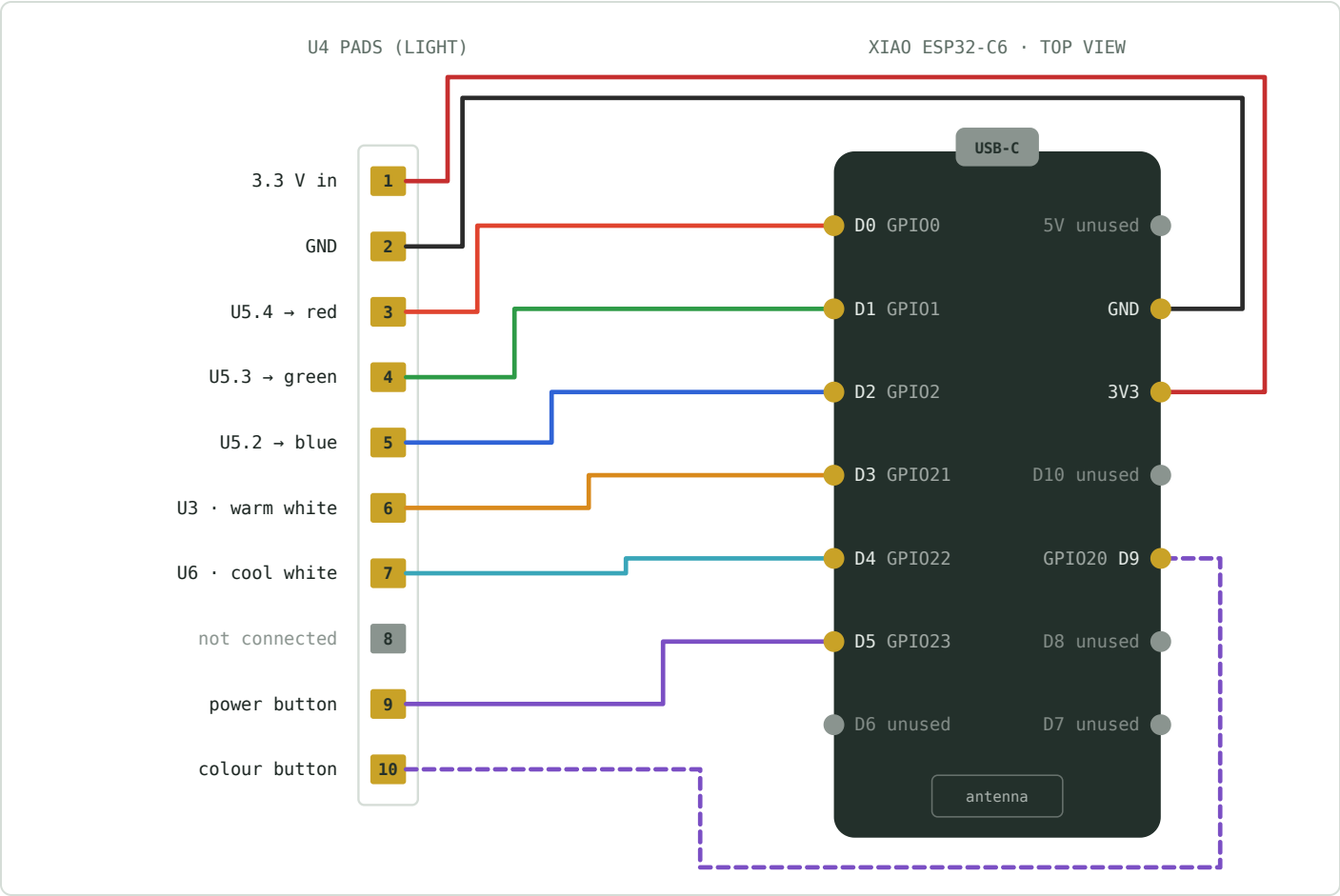
CYNC REVEAL HD+ LIGHT · U4 REPLACEMENT

Seeed XIAO ESP32-C6 in place of the JXC8720-18

Remove the Wi-Fi/BLE module at U4 and wire the XIAO to its ten pads. The light's three SM15133EH drivers, the buttons and the 3.3 V regulator stay as they are. Pin choices match `undercabinet-light.yaml`.

Wiring



Pad numbers are as you traced them: pad 1 is 3.3 V, pad 2 is GND. The dashed wire runs under the XIAO to reach D9 on the far header.

Wire list

WIRE	XIAO PIN	GPIO	U4 PAD	REACHES ON THE LIGHT	CONFIG SUBSTITUTION
Power	3V3	–	1	3.3 V from the light's regulator	–
Ground	GND	–	2	Ground	–
Red	D0	GPIO0	3	U5.4 DIM, red LEDs (154 Ω in series)	red_pin
Green	D1	GPIO1	4	U5.3 DIM, green LEDs	green_pin
Blue	D2	GPIO2	5	U5.2 DIM, blue LEDs	blue_pin
Warm white	D3	GPIO21	6	U3.2/3/4 DIM, all three channels	warm_white_pin
Cool white	D4	GPIO22	7	U6.2/3/4 DIM, all three channels	cool_white_pin
Power button	D5	GPIO23	9	Power button, 0 V when pressed	power_button_pin
Colour button	D9	GPIO20	10	Colour button, 0 V when pressed	colour_button_pin
No wire	–	–	8	Not connected on the light	–

None of these GPIOs are ESP32-C6 strapping pins (GPIO4, 5, 8, 9, 15). D6 and D7 are left free because the boot ROM drives UART0 on them; D8, D10 and the antenna-switch pins (GPIO3, GPIO14) are also left alone.

Fit-up and first power-on

> **No USB while wired.** The XIAO's own regulator and the light's 3.3 V supply would both drive the same rail. Flash it on the bench over USB first, then install it and update over the air.

- [ ] Flash `undercabinet-light.yaml` over USB with the XIAO off the light, and check it joins Wi-Fi and appears in Home Assistant.
- [ ] Remove the JXC8720-18 from U4 with hot air and clean the pads.
- [ ] Insulate the underside of the XIAO with Kapton tape. Its battery pads and test points must not touch the light's board.
- [ ] Keep the wires short. Place the XIAO so its antenna end overhangs the board edge where the old module's antenna was, away from the metal channel.
- [ ] On first power-up, watch for any channel that flashes during boot. If one does, add a 100 kΩ pull-down from that line to GND.
- [ ] Set the light to pure red, green and blue in Home Assistant and check each colour. The pin order was confirmed on the first light.
- [ ] Dim to 1% and raise `Minimum Brightness` until every channel glows evenly; it uses the same % as the brightness slider. `PWM Frequency` defaults to 4882 Hz, which showed no flicker on the first light.